

(29)

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SLOT: A

CO2-AT3

CLASS TEST

Machine Learning

1) Uses simple algorithms like Decision Trees, SVM

2) Requires small to medium datasets

3) Manual feature extraction

4) Faster to train

5) Works well on structured data

Deep Learning

1) Uses Multi-Layer Artificial Neural Networks

2) Requires very large datasets.

3) Automatic feature extraction.

4) Takes more training time and computational power

5) Excels on images, audio and text.

ML Applications:

1) Email Spam Detection.

2) Credit card fraud detection.

DL Applications:

1) Face Recognition.

2) Self-driving cars.

- 2) Explain the concept of Representation Learning.
Discuss how the width and depth of a neural network influence its learning capability?

Sol: Representation Learning: Representation learning is a technique where in which a neural networks automatically learns useful features from the raw data without manual feature engineering.

Effect of Width:

- * More neurons per layer.
- * Learns more patterns.
- * Increases computational cost and memory usage.

Effect of Depth:

- * More hidden layers.
- * Learns complex and hierarchical features.
- * Improves accuracy but increases training time and may cause vanishing gradients.

Sim Example:

In Image Recognition:

- * Layer 1 learns edges
- * Layer 2 learns shapes
- * Layer 3 learns object parts.
- * Final Layer identifies the complete object.
(eg: cat or car)

3) Compare RELU, Leaky RELU (LReLU) and ELU activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable cases.

Sol: Function: RELU

Formula:

$$f(x) = \max(0, x)$$

Advantage: Fast, simple, reduces vanishing gradient.

Disadvantage: Dead neuron problem.

use case: CNNs, images classification.

Function: Leaky RELU

Formula: $f(x) = x$ if $x > 0$ else αx

Advantage: solves dead neuron issue

Disadvantage: α must be chosen.

use case: Deep neural networks.

Function: ELU

Formula: x if $x > 0$, $\alpha(e^x - 1)$ if $x \leq 0$

Advantage: Faster convergence, better accuracy.

Disadvantage: More computation

use case: complex deep learning models.

4) Explain the concept of unsupervised learning in neural networks?

Sol: Unsupervised Learning: A learning method where the model learns patterns from unlabeled data.

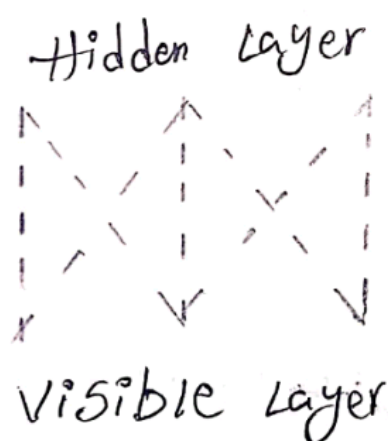
Restricted Boltzmann Machine (RBM)

- * Has visible and hidden layers.
- * Learns probability distribution of input data.
- * Used for feature extraction and recommendation systems.

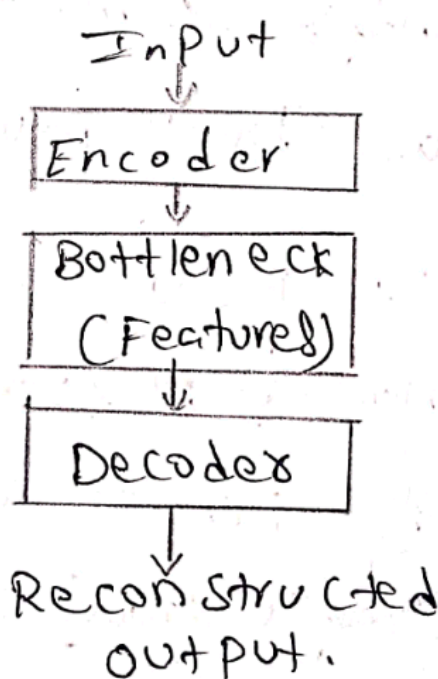
Autoencoder:

- * Consists of Encoder, Bottleneck and Decoder.
- * Compresses data and reconstructs it.
- * Used for dimensionality reduction, denoising and anomaly detection.

RBM



Autoencoder:



5) Explain the concept of Autoencoder and its uses in Deep learning?

Sol: Autoencoder: An Autoencoder is an unsupervised neural network that learns to compress input data into a lower-dimensional representation and then reconstruct it back to the original form.

Architecture:

- 1) Encoder
- 2) Bottleneck
- 3) Decoder

Uses:

- 1) Dimensionality reduction
- 2) Image denoising
- 3) Data compression
- 4) Anomaly detection
- 5) Feature extraction

Input Data

↓
Encoder

↓
Latent space
(compressed)

↓
Decoder

↓
Reconstructed
Output